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Methods

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Inference and Visualization of Complex Genotype–Phenotype Maps

Carlos Martí-Gómez [1]*, Juannan Zhou [2,3], Wei-Chia Chen [4], Arlin Stoltzfus [5], Justin B. Kinney [1], David M. McCandlish [1]*

[1]Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11724, USA
[2]Department of Biology, University of Florida, Gainesville, FL 32611, USA
[3]Genetics Institute, University of Florida, Gainesville, FL 32611, USA
[4]Department of Physics, National Chung Cheng University, Chiayi, Taiwan 62102, Republic of China
[5]Institute for Bioscience and Biotechnology Research, Rockville, MD 20850, USA
***Corresponding authors:** E-mails: martigo@cshl.edu; mccandlish@cshl.edu.
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Abstract

Understanding how biological sequences give rise to observable traits, that is, how genotype maps to phenotype, is a central goal in biology. Yet, our knowledge of genotype–phenotype maps in natural systems remains limited by the high dimensionality of sequence space and the context-dependent effects of mutations. The emergence of Multiplex assays of variant effect (MAVEs) and the availability of ever growing collections of natural sequences offer new opportunities to characterize these maps at an unprecedented scale. However, tools for statistical and exploratory analyses of such high-dimensional data are still needed. To address this gap, we developed *gpmap-tools* (https://github.com/cmarti/gpmap-tools), a *python* library integrating models for inference, phenotypic imputation, and error estimation from MAVE data or natural sequences in the presence of genetic interactions of any order. *gpmap-tools* also provides methods for summarizing patterns of epistasis across sites and visualization of genotype–phenotype maps with millions of genotypes. We demonstrate its utility by inferring genotype–phenotype maps containing 262,144 variants of the Shine-Dalgarno sequence, a key motif for mRNA translation in bacteria, from both genomic 5′UTR sequences and MAVE data. Visualization of the inferred landscapes consistently revealed high-fitness ridges that link core motifs at different distances from the start codon, motivating a new, highly interpretable thermodynamic model for this system. In summary, *gpmap-tools* provides a flexible, interpretable framework for studying complex genotype–phenotype maps, offering new insights into the architecture of genetic interactions and their evolutionary consequences.

Keywords genotype–phenotype map, fitness landscape, epistasis, Gaussian process, Shine-Dalgarno sequence

Introduction

The genotype–phenotype map describes how changes in biological sequences, such as DNA, RNA, or proteins, give rise to variation in observable traits. Understanding this relationship is essential across many areas of biology, ranging from evolutionary theory (Wright 1932; Kondrashov et al. 2002; Phillips 2008; Weinreich et al. 2013; De Visser and Krug 2014; Sailer and Harms 2017; Bank 2022; Johnson et al. 2023) and human disease (Moore and Williams 2009; Dasari et al. 2021; Moulana et al. 2023b), to synthetic biology, protein engineering applications (Yang et al. 2019; Freschlin et al. 2022; Lipsh-Sokolik and Fleishman 2024) and plant and animal breeding (De Los Campos et al. 2013; Sackton and Hartl 2016; Soyk et al. 2020; Dwivedi et al. 2024). However, our current understanding of genotype–phenotype maps in nature remains limited due to two fundamental challenges. First, the number of possible genotypes is astronomically large, making it impossible to explore more than a tiny fraction of this space in any empirical setting. Second, the phenotypic effect of a mutation often depends on the genetic background in which it occurs, a phenomenon known as epistasis (Starr and Thornton 2016; Posfai et al. 2018; Domingo et al. 2019; Miton et al. 2021; Bank 2022). Addressing this context-dependence requires experimental and computational approaches that are capable of capturing these complex genetic interactions.

One powerful approach to study empirical genotype–phenotype maps is to experimentally construct sequence variants and measure their functional consequences. Historically, these studies were limited by the difficulty of engineering a large number of genetic variants and quantifying their phenotypes at scale, restricting most empirical genotype–phenotype maps to small

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numbers of genotypes, typically ranging from tens to a few hundreds (Chou et al. 2011; Khan et al. 2011; Flynn et al. 2013; Szendro et al. 2013; Ogbunugafor et al. 2016; Weinreich et al. 2018; Gao et al. 2022; Aguirre et al. 2023; Zebell et al. 2025). The development of Multiplex assays of variant effects (MAVEs) (Kinney et al. 2010; Fowler and Fields 2014; Kinney and McCandlish 2019) has increased our phenotyping throughput by several orders of magnitude, enabling the simultaneous measurement of libraries containing thousands to millions of genotypes in a single experiment. These techniques have been used to characterize the phenotypic landscapes for short regulatory elements (Noderer et al. 2014; Bonde et al. 2016; Wong et al. 2018; Komarova et al. 2020; Kuo et al. 2020; Westmann et al. 2024, 2024b; Chattopadhyay et al. 2025; Kuo et al. 2025), RNAs (Domingo et al. 2018; Baeza-Centurion et al. 2019; Bendixsen et al. 2019; Soo et al. 2021; Rotrattanadumrong and Yokobayashi 2022) and proteins (O'Maille et al. 2008; Bank et al. 2016; Wu et al. 2016; Starr et al. 2017; Poelwijk et al. 2019; Jalal et al. 2020; Lite et al. 2020; Moulana et al. 2023; Papkou et al. 2023; Escobedo et al. 2025; Johnston et al. 2024; Sundar et al. 2024; Zarin and Lehner 2024; Herrera-Álvarez et al. 2025), as well as combinatorial gene interactions (Bakerlee et al. 2022). Yet, the highly combinatorial nature of these data poses significant challenges, and accurate inference typically relies on complex latent-variable models (Bloom 2015; Otwinowski et al. 2018; Tareen et al. 2022; Tonner et al. 2022; Faure and Lehner 2024) or neural networks (Bryant et al. 2021; Gelman et al. 2021; Sethi and Zhou 2025). Gaussian processes offer an alternative class of flexible models that both capture high-order interactions and provide accurate uncertainty quantification (Romero et al. 2013; Yang et al. 2019; Zhou and McCandlish 2020; Zhou et al. 2022, 2025; Petti et al. 2025). Moreover, the mathematical tractability of Gaussian processes means that they can be designed and interpreted in terms of existing genetic concepts. For example, empirical variance component regression (Zhou et al. 2022) explicitly learns the variance explained by epistatic interactions of different orders and uses this information for accurate inference and statistical analysis of complex genotype–phenotype maps from MAVE data (Zhou et al. 2022).

An alternative approach to study genotype–phenotype maps consists in analyzing collections of natural sequences. Since natural selection tends to preserve functional sequences, we can assume that the probability of observing a given sequence in nature depends on how well it performs its function. Thus, the probability distribution over sequences that perform a specific function can be interpreted as a genotype–phenotype map, where the phenotype is the probability of observing a sequence. Independent site models, such as Position-Weight Matrices (PWMs), estimate this sequence probability distribution by assuming that positions are independent from each other (Stormo 2013), whereas pairwise interaction models, also known as Potts models, relax this assumption by allowing interactions between pairs of positions (Morcos et al. 2011; Sly 2011; Ekeberg et al. 2013; Stein et al. 2015; Haldane and Levy 2021). These models have proven very effective in predicting structural contacts in proteins (Marks et al. 2012; Haldane et al. 2016, 2018) and between proteins (Bitbol et al. 2016; Malinverni and Babu 2023), as well as for predicting the effects of mutations in human proteins (Hopf et al. 2017). These types of pairwise interaction models have also been useful for identifying novel functional proteins (Russ et al. 2020) and regulatory sequences (Yeo and Burge 2004), and for quantifying the strength of selection at the gene level (Vigué and Tenaillon 2023). A recently proposed Gaussian process model called SeqDEFT further generalizes pairwise interaction models and allows inference of complex genotype–phenotype maps with higher-order epistatic interactions using readily available collections of natural sequences (Chen et al. 2021, 2024).

Another important challenge is the interpretation of complex genotype–phenotype maps. One way to develop an intuitive understanding of complex datasets is through data visualization tools. An approach to visualizing empirical genotype–phenotype maps is to embed the Hamming graph representing it, where each node is a genotype and edges represent single-point mutations, into a low-dimensional space. For instance, one can embed the graph by placing genotypes according to their Hamming distance to a reference sequence on one axis and their phenotype on the other (Wright 1932; Brouillet et al. 2015; Domingo et al. 2018; Baeza-Centurion et al. 2019; Bendixsen et al. 2019; Escobedo et al. 2025), or applying spectral and force-directed layouts (Starr et al. 2017; Fragata et al. 2019; Martin and Ahnert 2022; Herrera-Álvarez et al. 2025). However, these representations are not directly related to the conceptual framework of fitness peaks, valleys, and plateaus that has shaped much of our theoretical understanding of genotype–phenotype maps (Wright 1932; de Visser et al. 2018). A different strategy is to construct a low-dimensional representation that reflects the evolutionary dynamics induced by the genotype–phenotype map of interest e.g. where the distances between genotypes represent the expected time to evolve between them for a population under selection for high phenotypic values (McCandlish 2011). This property is very useful because it places sets of functional sequences that are inaccessible to each other i.e. peaks, far apart in the visualization, naturally displaying the key genetic interactions separating them, i.e. valleys. This technique has been successfully applied to uncover qualitative features of different genotype–phenotype maps (Zhou and McCandlish 2020; Chen et al. 2021; Zhou et al. 2022; Weinstein et al. 2023; Avizemer et al. 2025), as well as to study how the structure of the genetic code influences protein evolution (Rozhoňová et al. 2024), illustrating its potential as a general framework for interpreting and comparing complex genotype–phenotype relationships.

Here, we present *gpmap-tools*, a *python* library for studying empirical genotype–phenotype maps with genetic interactions of every possible order. In particular, *gpmap-tools* allows Gaussian process inference of large genotype–phenotype maps with different types and magnitudes of epistasis using both MAVE and natural sequence diversity data. Moreover, it enables calculation of the variance explained by genetic interactions involving specific sets of sites and statistical analysis of quantities of interest, such as mutational effects and epistatic coefficients, providing a powerful tool to characterize the structure and complexity of genetic interactions even in the presence of missing data and experimental noise. Finally, *gpmap-tools* provides methods for visualizing genotype–phenotype maps, investigating the sequence features that distinguish different regions of the representation, and accelerated rendering of plots containing millions of genotypes, enabling exploration of complex genotype–phenotype maps at unprecedented scale and resolution.

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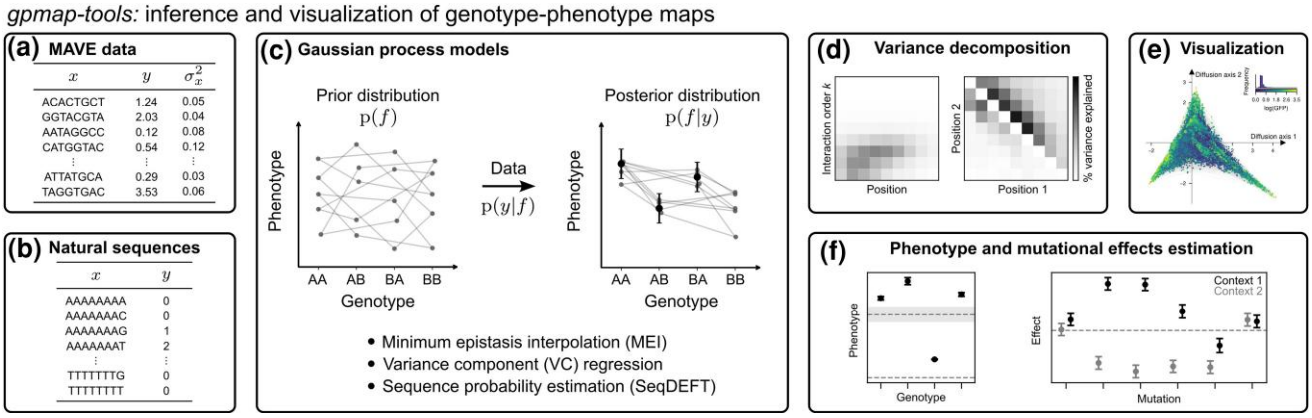

Fig. 1. Overview of the functionality provided by *gpmap-tools*. a, b) The software uses data from Multiplex Assays of Variant Effects (MAVEs) (a) or natural sequence variation (b) as input. c) *gpmap-tools* infers empirical genotype–phenotype maps using Gaussian process models by combining a prior distribution over the set of possible genotype–phenotype maps with likelihood functions for the two data types, yielding a posterior distribution that reflects our updated knowledge of the genotype–phenotype map given the data. d–f) *gpmap-tools* enables interpretation of the inferred maps by decomposing the overall phenotypic variance into components associated with interactions of different orders and specific subsets of sites (d), visualizing the full genotype–phenotype landscape (e), and computing the posterior distributions for specific genotypes or mutational effects of interest (f).

We demonstrate the capabilities of *gpmap-tools* by inferring the fitness landscape of the Shine-Dalgarno (SD) sequence from two fundamentally different types of data: (i) natural sequence diversity in genomic 5′ untranslated regions (UTRs) and (ii) MAVE data (Kuo et al. 2020). The inferred landscapes reveal a shared structure consisting of peaks corresponding to 16S rRNA binding at different distances relative to the start codon. These peaks are connected by extended ridges of functional sequences when the corresponding binding sites are spaced three nucleotides apart. These ridges arise from overlapping SD motifs in the same sequence, a consequence of the motif's quasi-repetitive structure, which allows new binding sites to emerge at offset positions without disrupting functionality. Building on this qualitative understanding of the genotype–phenotype map, we fit a simplified mechanistic model with parameters that have clear biophysical interpretations. This model allows us to disentangle the effects of mutations on binding at different registers relative to the start codon *in vivo*, while capturing the key structural features of the empirical landscape. Taken together, our analysis illustrates how *gpmap-tools* enables the inference and characterization of genotype–phenotype maps from diverse data sources, facilitating the discovery of simple molecular mechanisms capable of generating the observed architecture of epistatic interactions and offering insights into the evolutionary consequences induced by these complex genotype–phenotype maps.

Approach

In this section, we provide a brief technical overview of the methods for inference and interpretation of genotype–phenotype maps implemented in *gpmap-tools* (Fig. 1), for application see Results.

A genotype–phenotype map is a function that assigns a phenotype, typically a scalar value, to every possible sequence of length ℓ on α alleles (where e.g. $\alpha = 4$ for DNA and $\alpha = 20$ for proteins). This function can be represented by an α^ℓ-dimensional vector f containing the phenotype for every possible genotype. We begin by discussing several methods for quantifying the amount and type of epistasis present in any particular vector f.

Epistasis in genotype–phenotype maps

gpmap-tools implements two different methods for measuring the amount and pattern of epistasis in a given genotype–phenotype map: one based on the typical magnitude of local epistatic coefficients across all possible subsets of mutations and the other based on the proportion of phenotypic variance explained by genetic interactions of different orders or involving specific subsets of sites.

Local epistatic coefficients

The traditional epistatic coefficient quantifies how much the effect of a mutation $A \to a$ in one site changes in the presence of an additional mutation $B \to b$ in a second site in an otherwise identical genetic background C:

$$\epsilon = (f_{aBC} - f_{ABC}) - (f_{abC} - f_{AbC}).$$

The average squared epistatic coefficient $\overline{\epsilon^2}$ across all possible pairs of mutations and genetic backgrounds provides a measure of the variability in mutational effects between neighboring genotypes across the whole genotype–phenotype map (Zhou and McCandlish 2020). $\overline{\epsilon^2}$ can be efficiently calculated as a quadratic form $\overline{\epsilon^2} = \frac{1}{s} f^T \Delta^{(2)} f$, where s is the number of epistatic coefficients and $\Delta^{(2)}$ is a previously described sparse $\alpha^\ell \times \alpha^\ell$ positive semi-definite matrix (Zhou and McCandlish 2020). This statistic can be generalized to characterize the typical size of local P-way epistatic interactions (Chen et al. 2021) and to a setting in which alleles are naturally ordered e.g. copy number (Chen et al. 2024). In practice, to quantify how locally epistatic a given genotype–phenotype map is, we compute $\sqrt{\overline{\epsilon^2}}$. This quantity can be interpreted in absolute terms, as it has the same units as the phenotype, or it can be interpreted relative to the typical magnitude of mutational effects by comparing it with the root mean squared mutational effect.

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Variance components

A genotype–phenotype map f can be decomposed into the contributions of $\ell + 1$ orthogonal subspaces $f = \sum_k f_k$, where f_k represents a function containing epistatic interactions solely of order k. These orthogonal components f_k are obtained by projecting f onto the kth order subspace using the projection matrix P_k. This decomposition enables quantification of the variance explained by interactions of different orders, providing a global summary of the complexity of genetic interactions in a genotype–phenotype map (Happel and Stadler 1996; Stadler 1996; Stadler and Happel 1999; Stadler 2002; Zhou et al. 2022).

Here, we show that each f_k can be further decomposed into the contribution of $\binom{\ell}{k}$ smaller orthogonal subspaces $f_k = \sum_{U:|U|=k} f_U$, where f_U represents a function containing genetic interactions only among the k sites in U. These orthogonal components f_U are obtained by projecting the function f into the corresponding subspace using the orthogonal projection matrix P_U given by:

$$P_U(x, x') = \alpha^{-\ell} \prod_{\substack{p \in U \\ x_p = x'_p}} (\alpha - 1) \prod_{\substack{p \in U \\ x_p \neq x'_p}} (-1),$$

for any pair of sequences x, x' (see Supplementary Information A). A similar decomposition was shown for certain models of sequence–function relationships parametrized using a specific scheme of interaction terms (Park et al. 2024; Posfai et al. 2025), but this formulation allows the direct decomposition of the function f without relying on a particular parameterization.

These projection matrices allow us to quantify not only the variance explained by interactions of different orders but also the variance explained by all possible subsets of sites U. Although there are 2^ℓ such subsets in total, they can be aggregated to yield informative, low-dimensional summary statistics. For instance, we can compute the variance explained by order-k epistatic interactions involving a specific site or pair of sites, or the variance explained by interactions of all orders involving each pair of sites (see Supplementary Information B) (Crawford et al. 2017; Reddy and Desai 2021). Through these capabilities, *gpmap-tools* enables a fine-grained decomposition of epistatic variance, offering new insights into the structure and complexity of genetic interactions across sites (Fig. 1d).

Gaussian process inference of genotype–phenotype maps

Gaussian process models are a class of Bayesian nonparametric models that place a multivariate Gaussian prior distribution over all possible functions and compute the posterior distribution given observed data (Rasmussen and Williams 2008). In our case, we assign a zero-mean Gaussian prior distribution over genotype–phenotype maps $\mathrm{p}(f)$ characterized by either its covariance matrix K or precision matrix C. The covariance matrix K is most often defined through a kernel function that returns the prior covariance between any pair of sequences. Then, given some data y and using a likelihood function $\mathrm{p}(y \mid f)$, we update the probability distribution of plausible genotype–phenotype maps to be consistent with these observations by computing the posterior distribution $\mathrm{p}(f \mid y)$ (Fig. 1c).

Interpretable priors

gpmap-tools implements two families of priors based on the two approaches to quantify epistasis in genotype–phenotype maps described above: one family that is defined in terms of local epistatic coefficients and a second that is defined in terms of variance components. The first prior is parametrized by its precision matrix $C = \frac{a}{s}\Delta^{(P)}$ and assigns a prior probability to f depending on its average squared epistatic coefficient of order P, i.e. $\log \mathrm{p}(f \mid a) \propto -\frac{a}{2s} f^T \Delta^{(P)} f$ (Zhou and McCandlish 2020; Chen et al. 2021). This prior implicitly leaves genetic interactions of order $k < P$ unconstrained, and hence correspond to the use of an improper Gaussian prior. For examples, for $P = 2$ additive effects are not penalized, for $P = 3$ additive and pairwise effects are not penalized, etc. For fixed P, this family of priors has a single hyperparameter a that is inversely proportional to the expected average squared local P-epistatic coefficient under the prior. As $a \to 0$, we assign the same prior probability to every possible genotype–phenotype map. On the other hand, as $a \to \infty$, we decrease the prior probability of genotype–phenotype maps with nonzero local P-epistatic coefficients (Chen et al. 2021).

The second family of priors are the variance component priors, which are parametrized by their covariance matrix $K = \sum_{k=0}^{\ell} \lambda_k K_k$, where K_k represents the covariance matrix for genotype–phenotype maps with only kth-order interactions. The $\ell + 1$ hyperparameters λ_k control the variance explained by genetic interactions of order k (Neidhart et al. 2013) and equivalently the decay in the predictability of mutational effects and epistatic coefficients as the number of mutations separating two genetic backgrounds increases (Zhou et al. 2022). The formal relationship between the two sets of priors is that the priors based on the $\Delta^{(P)}$ operators can be obtained as limits of the variance component prior (Zhou et al. 2022).

These prior distributions for f have hyperparameters with clear biological interpretations in terms of the expected magnitude and type of epistasis. This allows users to define the corresponding priors in a principled and interpretable way. Moreover, under the assumption that the structure of epistasis observed in the data generalizes to the full genotype–phenotype map, *gpmap-tools* can infer these hyperparameters using either cross-validation or kernel alignment (Rasmussen and Williams 2008; Wang et al. 2015), providing estimates that are both data-driven and biologically meaningful.

Likelihood functions

gpmap-tools implements two likelihood functions for inference of genotype–phenotype maps from different types of data. For experimental data, the vector y contains the measurements associated to a subset of sequences x with known Gaussian measurement variance σ_x^2 (Fig. 1a). Thus, the likelihood function is given by

$$\mathrm{p}(y \mid f) = \mathrm{N}(f_x, D_{\sigma_x^2}),$$

where $D_{\sigma_x^2}$ is a diagonal matrix with σ_x^2 along the diagonal.

For observations of natural sequences, data consist of the number of times N_i a given sequence i was observed out of a total

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of $N_T = \sum_i N_i$ observations (Fig. 1b). In this case, the likelihood function is given by the multinomial distribution

$$\mathrm{p}(N \mid \pi, N_T) = \mathrm{Multinomial}(\pi, N_T),$$

where π is the vector representing the sequence probability distribution, such that π_i corresponds to the probability of observing sequence i.

Posterior distributions

gpmap-tools enables the computation of the posterior distribution over the space of possible genotype–phenotype maps using both Gaussian and multinomial likelihood functions. Under a Gaussian likelihood, the posterior distribution is also a Gaussian $\mathrm{p}(f \mid y, D_{\sigma^2}) = \mathrm{N}(\hat{f}, \Sigma)$ with closed form analytical solutions for the mean $\hat{f}$ and covariance matrix Σ. *gpmap-tools* implements the classical solution expressed in terms of the prior covariance matrix K (Rasmussen and Williams 2008) but also the solution when the prior is defined by its precision matrix C, which are equivalent when $C = K^{-1}$ (see Supplementary Information F).

Under non-Gaussian likelihood functions, such as the multinomial likelihood, the posterior distribution has no closed form analytical solution. However, given that $\log \mathrm{p}(\mathrm{f} \mid \mathrm{y})$ is proportional to $\log \mathrm{p}(\mathrm{y} \mid \mathrm{f}) + \log \mathrm{p}(\mathrm{f})$, which can be efficiently computed for any f, *gpmap-tools* leverages optimization methods to find the maximum a posteriori (MAP) $\hat{f}$ and compute an approximate Gaussian posterior using the Laplace approximation, where the posterior covariance matrix Σ is defined by the inverse Hessian of the posterior at its mean $\hat{f}$ (Rasmussen and Williams 2008).

These solutions are completely general, in the sense that they hold for arbitrary valid prior covariance of precision matrices. However, as we will explain below, *gpmap-tools* implements highly optimized versions of these calculations that take advantage of the structure of sequence space and our specific choices for C and K.

Inference of genotype–phenotype maps with *gpmap-tools*

gpmap-tools combines prior distributions with likelihood functions into a number of Gaussian process models for inference of complete genotype–phenotype maps.

Minimum epistasis interpolation

The minimum epistasis interpolation (MEI) method was originally proposed in terms of finding the f_z at unobserved sequences z given the known phenotype f_x at sequences x by minimizing the average squared epistatic coefficient $\overline{\epsilon^2}$ over the complete genotype–phenotype map (Zhou and McCandlish 2020). *gpmap-tools* provides a generalization to local epistatic coefficients of any order P, (by minimizing $f^T \Delta^{(P)} f$ Chen et al. 2021), incorporates known Gaussian measurement noise through σ_x^2 and, by re-framing minimum epistasis interpolation as a Gaussian process model, enables uncertainty quantification via the posterior covariance (see Supplementary Information E,F).

Empirical variance component regression

Empirical variance component regression (VC regression), proposed in (Zhou et al. 2022), combines a variance component prior parameterized by the variance λ_k associated to interactions of each possible order k with a Gaussian likelihood with known noise variance σ_x^2 to compute the exact Gaussian posterior distribution over f. The hyperparameters λ_k controlling the expected variance explained by interactions of order k under the prior are optimized through kernel alignment (Wang et al. 2015). This procedure minimizes the squared distance between the covariance under the prior and the empirical distance-covariance function computed from the incomplete data. While a naive kernel alignment implementation requires computation with large covariance matrices, the prior covariance between two sequences depends only on the Hamming distance between them, resulting in only $\ell + 1$ different values. As a consequence, the problem of kernel alignment can be efficiently solved as a simpler $\ell + 1$-dimensional constrained weighted least squares problem (Zhou et al. 2022).

Sequence probability distribution estimation

gpmap-tools also implements the SeqDEFT method for estimating probability distributions π over sequence space (Chen et al. 2021). SeqDEFT parametrizes the probability distribution as

$$\pi_i = \frac{e^{-\phi_i}}{\sum_j e^{-\phi_j}}$$

and defines an improper prior distribution over the latent phenotype ϕ that penalizes local epistatic coefficients of order P given by $\log \mathrm{p}(\phi \mid a) \propto -\frac{a}{2s} \phi^T \Delta^{(P)} \phi$, which is combined with a multinomial likelihood function to compute an approximate posterior distribution over ϕ.

The hyperparameter a is optimized by maximizing the cross-validated log-likelihood under the MAP estimate over a 1D grid search. This also enables us to examine the behavior of the model towards the two limiting solutions i.e. when local epistatic coefficients are unconstrained ($a = 0$) or forced to be zero ($a \to \infty$) (Chen et al. 2021).

Visualization of genotype–phenotype maps

Genotype–phenotype maps are inherently high-dimensional objects, and thus difficult to visualize in an intuitive manner. *gpmap-tools* implements a previously proposed strategy for visualizing fitness landscapes (McCandlish 2011) that computes embedding coordinates for genotypes such that squared distances between pairs of genotypes in the low-dimensional representation approximate the expected times to evolve from one to another under selection for high phenotypic values (Fig. 1e). This layout highlights regions of sequence space containing highly functional genotypes that are nevertheless poorly accessible to each other e.g. fitness peaks separated by valleys, or sets of sequences where the intermediates are functional but the order of the intervening mutations is highly constrained.

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Evolutionary model

We assume a weak mutation model of evolution in haploid populations, such that mutations are always fixed or lost before a new mutation arises (McCandlish 2011, 2018; Zhou and McCandlish 2020; Chen et al. 2021). Under this model, the evolutionary rate $Q(i, j)$ from genotype i to j depends on the mutation rate $M(i, j)$ (which we assume is taken from a time-reversible mutational model) and the probability of fixation relative to a neutral mutation (Bulmer 1991; McCandlish and Stoltzfus 2014):

$$Q(i,j) = \begin{cases} M(i,j)\frac{S(i,j)}{1-e^{S(i,j)}} & \text{if } i \text{ and } j \text{ are neighbors} \\ -\sum_{k\neq i} Q(i,k) & \text{if } i = j \\ 0 & \text{otherwise,} \end{cases}$$

where $S(i, j)$ is the scaled selection coefficient of genotype j relative to genotype i. For the purposes of constructing a useful visualization, we then assume that this scaled selection coefficient is proportional to the phenotypic differences between the two genotypes i.e. $S(i, j) = c(f(j) - f(i))$, where the constant c can be interpreted as the scaled selection coefficient ($2N_e s$, for a Haploid Wright-Fisher population) associated with a phenotypic difference of 1. Unless specifically studying the role of mutational biases on evolution on empirical landscapes, we would typically assume that $M(i, j) = 1$ for any i, j pair (i.e. measuring time in units of the inverse mutation rate), and focus on the evolutionary dynamics induced by the structure of the genotype–phenotype map alone. This model assigns a low but nonzero probability of fixation to deleterious mutations and has a unique stationary distribution $\pi(i)$ given by

$$\pi(i) = \frac{\pi_M(i)e^{cf(i)}}{\sum_j \pi_M(j)e^{cf(j)}},$$

where $\pi_M(i)$ are the time-reversible neutral stationary frequencies, which are uniform in the absence of mutational biases (Sella and Hirsh 2005; McCandlish et al. 2015). The stationary distribution can be used to select a reasonable value of c for our evolutionary process. When representing a probability distribution, such as one inferred using SeqDEFT, setting $f(i) = \log \pi(i)$ and $c = 1$ will result in a stochastic process in which the stationary distribution exactly matches the estimated genotype probabilities, providing a very natural representation of the landscape. When inferring the genotype–phenotype map from MAVE data, c can be adjusted so that the mean phenotype under the stationary distribution aligns with realistic natural values e.g. the phenotype associated to a wild-type or reference sequence(s). Alternatively, a range of c values can be used to generate a family of visualizations for a single genotype–phenotype map to reflect the evolutionary impact of its structure under different assumptions concerning the relative strengths of selection and drift.

Low-dimensional representation

The right eigenvectors r_k of Q associated to the largest eigenvalues λ_k ($\lambda_1 = 0 > \lambda_2 \geq \lambda_3 \geq \ldots$) can be computed using iterative methods that leverage the sparse structure of Q. When appropriately normalized and re-scaled as $u_k = \frac{1}{\sqrt{-\lambda_k}}\frac{r_k}{r_k^T D_\pi r_k}$, the first few r_k for $k \geq 2$ can be used as embedding coordinates, resulting in a low-dimensional representation in which squared distances between genotypes optimally approximate the commute times i.e. the sum of hitting times $H(i, j)$ from i to j and $H(j, i)$ from j to i, thus separating sets of functional genotypes that are largely inaccessible to each other for a population evolving under selection for high phenotypic values:

$$\sum_{k=2}(u_k(i) - u_k(j))^2 \approx H(i,j) + H(j,i).$$

The eigenvalues λ_k represent the rates at which the associated eigenvectors become less relevant for predicting evolutionary outcomes with time. The associated relaxation times $-\frac{1}{\lambda_k}$ have units of expected number of substitutions and allow us to identify components that decay slower than expected under neutral evolution, where we note that if all mutations occur at rate 1, the neutral relaxation time is given by the reciprocal of the minimum number of alleles across sites. Because u_k captures the $k-1$th strongest barrier to the movement of a population in sequence space, we refer to u_k as diffusion axis $k-1$ (see McCandlish 2011; Chen et al. 2021, for more details).

Rendering and visualization

In addition to computing the coordinates u_k, *gpmap-tools* provides functionality at both high and low levels to plot and render the visualizations of genotype–phenotype maps using different backend plotting libraries. This includes the standard plotting library in python, *matplotlib* (Hunter 2007), for generating highly customized visualizations, and *plotly* (Hossain 2026), for generating interactive 3D visualizations that display the sequence associated to each node when hovering the mouse over them. Moreover, as rendering large numbers of points and lines becomes limiting in large datasets, the *gpmap-tools* plotting library leverages the power of *datashader* (Bednar et al. 2022) for efficiently rendering plots containing millions of different elements, achieving close to an order of magnitude speed up for large genotype–phenotype maps (Fig. S1).

Efficient computation with *gpmap-tools*

We aim to study genotype–phenotype maps with a number of genotypes ranging from a few thousands up to millions. However, all of the described methods require computing with unreasonably large matrices of size $\alpha^\ell \times \alpha^\ell$. For instance, to study a genotype–phenotype map for 9 nucleotides, a naive implementation would need to build a $4^9 \times 4^9$ matrix requiring 512GB of memory using 64 bit floating point numbers and over 100 billion operations to compute matrix-vector products. While some of the necessary matrices are sparse e.g. $\Delta^{(P)}$ and Q, allowing efficient storage and computation (McCandlish 2011; Zhou and McCandlish 2020; Chen et al. 2021), other matrices e.g. P_U and K, are dense.

gpmap-tools circumvents these challenges using two strategies. First, we note that every matrix A with entries A_{ij} depending only on the Hamming distance between sequence i and j, such as $\Delta^{(P)}$ as well as the dense matrices P_k and K_k, can be expressed as an ℓ-order polynomial in the Laplacian of the Hamming graph L (Zhou et al. 2022). This enables efficient computation of matrix-vector products $Ab = \sum_i^\ell c_i L^i b$ by multiplying the vector b by L up to ℓ times, e.g. $L^2 b = L(Lb)$, and taking linear combinations of the results without explicitly building the possibly dense matrix A.

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gpmap-tools also implements L as a linear operator (see Supplementary Information D). While this new linear operator-based implementation achieves comparable efficiency to a sparse matrix formulation in nucleotide space, it is an order of magnitude faster in protein spaces (Fig. S2). More importantly, the L linear operator requires virtually no time for construction and has much lower memory requirements (Fig. S2).

Second, we note that many of the relevant matrices can be obtained as ℓ-Kronecker products of $\alpha \times \alpha$ matrices, such as $P_U = \bigotimes_p^{\ell} P_p$. By using *scipy*'s (Virtanen et al. 2020) LinearOperators functionality, we can leverage the mixed Kronecker matrix-vector product property to enable computation of e.g. $P_U b$ without constructing P_U (see Supplementary Information C, Fig. S3). Rather than calculating explicit inverse matrices, we can likewise use these linear operators to find numerical solutions to matrix equations using Conjugate Gradient (CG). By combining multiple linear operators, we are able to compute the posterior variance for a small number of sequences of interest or the posterior covariance for any set of linear combinations of phenotypic outcomes e.g. calculating posterior variance for mutational effects in specific genetic backgrounds and epistatic coefficients of any order (Fig. 1f), while limiting the number of linear systems to solve with CG to the number of linear combinations of interest.

Results

In this section, we illustrate the power of *gpmap-tools* by studying the genotype–phenotype map of the SD sequence. The SD sequence is a motif located in the 5′UTR of most prokaryotic mRNAs. This motif is recognized by the 3^{prime} tail of the 16S rRNA through base pair complementarity with a region known as the anti Shine-Dalgarno (aSD) sequence, promoting translation initiation (Shine and Dalgarno 1975). Understanding how the SD sequence modulates protein translation *in vivo* is key for understanding gene regulation and in synthetic biology applications (Salis et al. 2009; Gilliot and Gorochowski 2024). Previous studies used existing sequence diversity (Hockenberry et al. 2018; Wen et al. 2020) and MAVE data (Bonde et al. 2016; Kuo et al. 2020) to build models for this genotype–phenotype map. However, these models cannot account for higher-order genetic interactions and provide limited understanding of the structure of the genotype–phenotype map. Thus, *gpmap-tools* offers a new opportunity to model and understand the patterns of genetic interactions and the main qualitative features that define this important regulatory element.

Inferring the probability distribution of the Shine-Dalgarno sequence

In this section, we infer a genotype–phenotype map of the SD sequence using observation of natural sequences from the 5′ untranslated regions (UTRs) across the whole *E. coli* genome. Here, we assume that the probability with which a sequence is observed is related to how well it performs its function, in this case, promoting translation initiation. To do so, we first extracted the 5′UTR sequences from 5,311 annotated genes and aligned them with respect to the start codon. Figure 2a shows

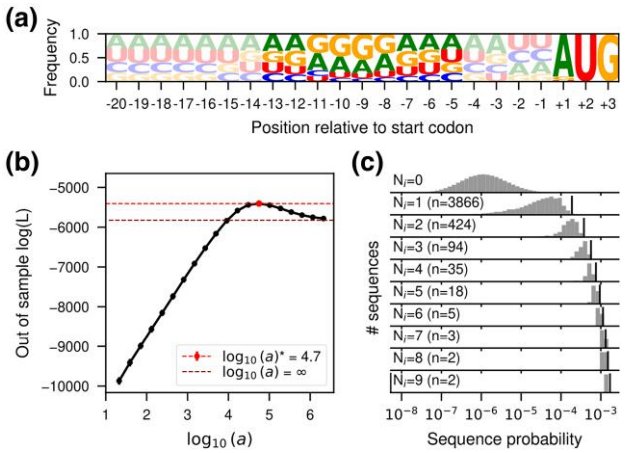

Fig. 2. Inference of the probability distribution of the Shine-Dalgarno sequence. a) Sequence logo representing the site-specific allele frequencies of 5,311 5′UTRs in the *E. coli* genome aligned with respect to the annotated start codon. The start codon and the nine nucleotide region four bases upstream are highlighted to emphasize the region most relevant for translation initiation. b) Log-likelihood computed in the 20% held-out sequences in 5-fold cross-validations of a series of SeqDEFT models ($P = 2$) under varying values of the hyperparameter a. The horizontal dashed lines represent the log-likelihood in the held-out data of the limiting maximum entropy model, corresponding to the independent sites model shown in (a), or the best SeqDEFT model maximizing the log-likelihood. c) Distribution of inferred sequence probabilities depending on the number of times N_i they were present in the *E. coli* genome represented in a logarithmic scale. Vertical black lines represent the empirical frequency N_i/N_T corresponding to each N_i value.

site-specific allele frequencies for up to 20bp upstream of the start codon. These allele frequencies showed a slight enrichment in purines between positions -13 and -5. This AG-rich pattern is characteristic of the location of the SD sequence, which classically exhibits the consensus sequence AGGAGGU complementary to the aSD sequence ACCUCCU (Shine and Dalgarno 1975; Hockenberry et al. 2018; Wen et al. 2020). Thus, from here onward, we focus on this 9 nucleotide region. We observed only 3, 690 out of the total $4^9 = 262,144$ possible 9-nucleotide sequences, most of which were observed only a single time. Given that the total number of observed sequences is two orders of magnitude smaller than the number of possible sequences, we expect many highly functional sequences to be unobserved.

Next, we used SeqDEFT to infer the probability distribution from which natural sequences are drawn. SeqDEFT enables accurate inference of sequence probabilities by defining a prior distribution parametrized by a, which controls how much the effects of mutations differ across neighboring genetic backgrounds. To estimate the value of the hyperparameter a, we fit models for increasing a values and evaluate their performance using cross-validation. Figure 2b shows that the SeqDEFT model defined by a^*, the value of a that maximizes the likelihood of the held-out data, better predicts the frequencies of held-out sequences than either the site-independent model (the maximum entropy model given by the site-independent frequency profiles, $a = \infty$, Fig. 2a) or the model determined by the empirical frequencies ($a \to 0$, which maximizes the likelihood). The fact that the optimum a^* is at an intermediate value of a provides strong support for the presence of epistatic interactions (Chen et al. 2021).

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We then estimated the SD sequence probability distribution under the prior defined by a^*, this time using all available data. To illustrate the behavior of the model, we compared the inferred probabilities using SeqDEFT with their observed empirical frequencies among *E. coli* 5'UTRs (Fig. 2c). Sequences that are observed more than 2 times are always inferred to be highly functional (i.e. high probability). However, there is a wide range of variability in the inferred probabilities of unobserved sequences, ranging 4 orders of magnitude. Many unobserved sequences even have larger probabilities than some sequences that are observed once, confirming our expectation that many highly functional sequences are not observed due to limited sampling. SeqDEFT enables these inferences by taking into account how similar the effects of mutations across different genetic backgrounds appear to be and incorporating this information into the prior. Finally, in order to quantify the overall magnitude of local epistatic interactions in the inferred landscape, we computed the root mean square local epistatic coefficient (0.32). This quantity is slightly lower than half of the size of the root mean squared mutational effect (0.78), indicating that adding a single mutation to the genetic background often substantially changes the effects of other mutations.

Inferring the genotype–phenotype map of the Shine-Dalgarno sequence from MAVE data

We next used data from a previously published MAVE (Kuo et al. 2020) measuring the expression of a GFP reporter controlled by a sequence library containing nearly all 262,144 possible nine nucleotide sequences spanning positions -13 to -5 relative to the start codon, i.e. the same region considered in our previous analysis of genomic 5'UTR sequences. We first used minimum epistasis interpolation (MEI) to estimate the phenotype for all missing genotypes by minimizing the total amount of local epistasis in the landscape. The estimated genotype–phenotype map had a root mean squared epistatic coefficient of 0.21. This quantity is comparable to the root mean squared size of mutational effects (0.26), indicating that there is substantial variability in the effects of mutations across neighboring genotypes. In fact, this shows that the variability in the effects of mutations across neighboring genetic backgrounds, relative to the average magnitude of mutational effects, is larger in the genotype–phenotype map inferred from MAVE data using MEI than when inferred from sequence diversity using SeqDEFT.

To better capture this high degree of epistasis, we turned to VC regression. We first characterized the magnitude of epistatic interactions present in the data in more detail by computing the correlation in the measured phenotypes between pairs of sequences depending on the number of mutations separating them. We found that these empirical correlations decayed quite quickly with Hamming distance e.g. pairs of sequences separated by three mutations only showed a correlation of 0.25 between their measured phenotypes (Fig. 3a), illustrating the high phenotypic unpredictability induced by extensive epistatic interactions. We then used the empirical correlations to estimate the variance component prior distribution under which the expected correlations most closely match the empirical values. Figure 3b shows the variance explained by interactions of every possible order (black) and the cumulative variance explained by interactions up to order k (gray) under this prior. The additive and pairwise component explained only 57.6% of the overall phenotypic variance (Fig. 3b, gray), suggesting an important influence of higher-order genetic interactions.

Once we estimated these variance components, we then inferred the complete genotype–phenotype map under the corresponding prior. These estimates recapitulated the experimental data remarkably well ($R^2 = 0.94$, Fig. 3c) and made predictions almost as accurate in held-out test sequences ($R^2 = 0.87$, Fig. 3d). Importantly, our estimates of the uncertainty of the phenotypic predictions are well calibrated, as we find approximately the expected fraction of measurements in the test set within posterior credible intervals (Fig. S4C). Comparing the predictive performance of MEI against VC regression as a function of the number of sequences used for training, we find that while the two models perform comparably well when the genotype–phenotype map is densely sampled, and MEI performs better with extremely low sampling (likely due to error in the estimation of variance components), overall VC regression exhibited substantially higher performance across a wide range of training data densities (Fig. S4A,B) and better calibration of the prediction's uncertainty (Fig. S4C).

Position-specific contributions to epistasis

An important advance in *gpmap-tools* is its ability to calculate the contribution of each site to genetic interactions of different orders (Supplementary Information B). Figure 3e shows this analysis for the genotype–phenotype map estimated with VC regression. Some positions i.e. -6 and -5, have an overall weak influence in the measured translational efficiency (the additive contributions of each site explain, on average, 0.61% of the total variance, whereas second- and third-order interactions involving either site explain an average of 1.53 and 2.83% of the total variance, respectively). In contrast, sites -13 to -10 have both strong additive and epistatic contributions (each of them has, on average, an additive and pairwise contribution of 7.32 and 8.73%, and interactions of order 3 and 4 involving each site explained on average 8.99 and 4.38% of the total variance, respectively). Interestingly, not all sites involved in higher-order interactions also have strong additive and pairwise contributions. For instance, sites -9 to -7 influence the phenotype more strongly through higher-order epistatic interactions (on average they had 2.60, 4.81, 8.44, and 5.16% of the variance explained by interactions of order 1 to 4 involving each site). Thus, we find that sites in the SD sequence have very heterogeneous contributions to genetic interactions of different orders, with some sites having stronger additive and lower order epistatic interactions, whereas other sites influence translation primarily via higher-order interactions.

These variances can be further decomposed into variances explained by epistatic interactions of any order involving each possible pair of sites (Supplementary Information B). This decomposition reveals that pairwise interactions are largely confined to sites within three nucleotides of each other in the primary sequence (interactions between a pair of positions within and beyond three nucleotides of each other explain an average of 5.11% and 1.11% of the pairwise variance, respectively) and are strongest between positions -13 to -10

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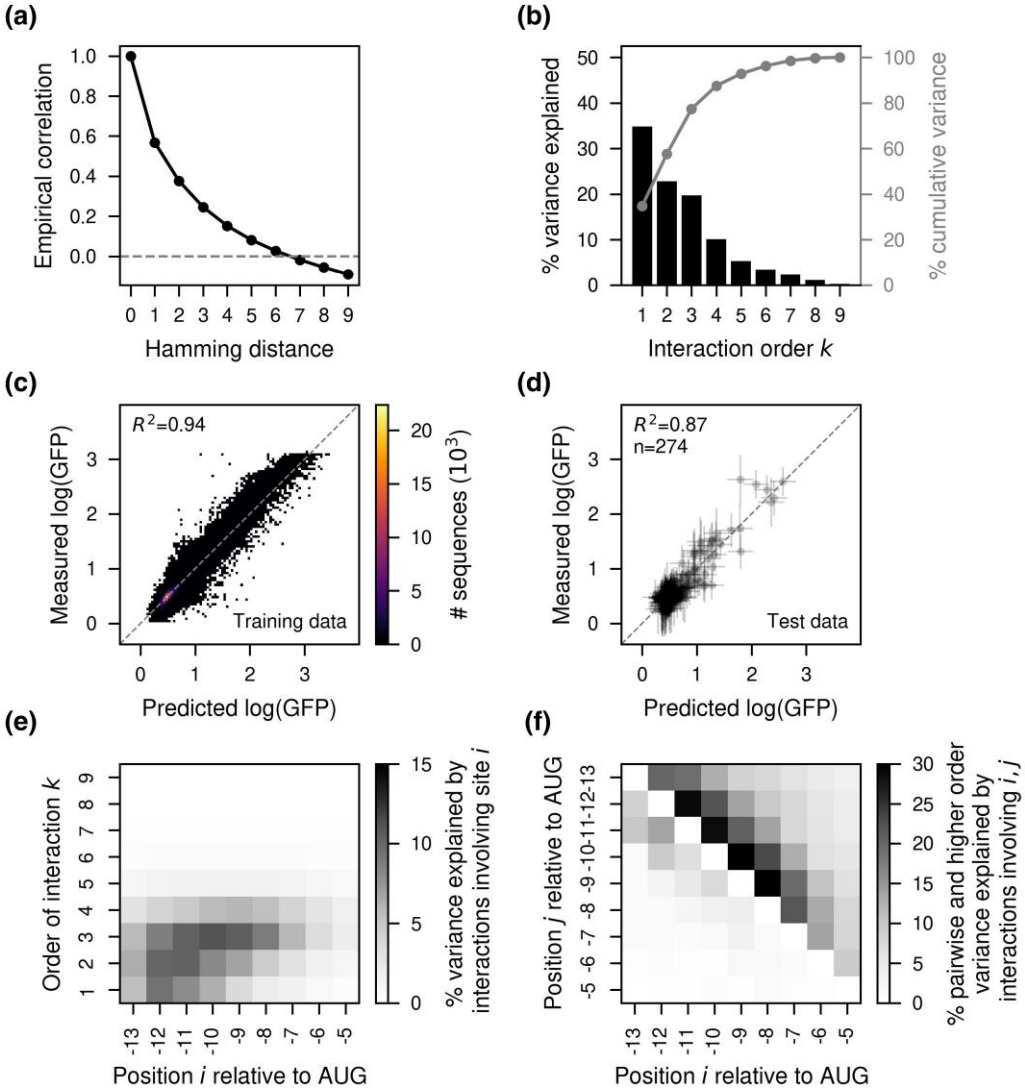

Fig. 3. VC regression analysis of the experimentally measured genotype–phenotype map for the Shine-Dalgarno sequence in the dmsC gene context (Kuo et al. 2020). a) Empirical distance-correlation function using the measured log(GFP) values in the experimentally evaluated sequences. b) Percentage of variance explained by interactions of order *k* in the inferred VC regression prior. Gray lines represent the cumulative percentage of variance explained by interactions up to order *k*. c) 2D histogram showing the comparison of the measured log(GFP) and the inferred values using VC regression for sequences used for model fitting. d) Comparison of the posterior distribution for held-out test sequences and the measured log(GFP) values. Horizontal error bars represent posterior uncertainty represented as the 95% credible interval, whereas vertical error bars correspond to the 95% confidence interval under each measurement's variance. e) Heatmap representing the percentage of variance explained by interactions of order *k* involving each position relative to the start codon. f) Heatmap representing the percentage of variance explained by pairwise (lower triangle) and higher-order (upper triangle) interactions that is explained by interactions involving pairs of positions relative to the start codon. See Supplementary Information B for details on calculating (e, f).

(Fig. 3f, lower triangle, average of 8.23%). This range of interaction is wider for higher-order interactions (higher-order interactions between a pair of positions within four nucleotides of each other explain on average 18.25% of the total variance explained by higher-order interactions) with the most prominent effects involving pairs of positions between -9 and -7 (Fig. 3f, upper triangle, higher-order interactions involving pairs within this window explain an average of 24.17% of the total variance explained by higher-order interactions). In contrast, interactions between sites separated by five or more nucleotides are rare across all orders (Fig. 3f). These findings indicate that the effect of a mutation at a given site depends primarily on nearby sites and becomes nearly independent of mutations beyond a 4-nucleotide range. Overall, the ability to quantify the variance explained by interactions of different orders and positional combinations offers a powerful framework for characterizing the nature and strength of epistasis, and for identifying communities of interacting sites within genotype–phenotype maps.

Visualizing the probability distribution of the SD sequence

In order to understand the main qualitative properties of this highly epistatic genotype–phenotype map, we generated a

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low-dimensional representation using our visualization technique. Figure 4a shows that the genotype–phenotype maps consists of at least three largely isolated peaks. These peaks correspond to the canonical SD motif AGGAG located at three consecutive positions relative to the start codon (-12, -11, -10), with a fourth central peak corresponding to the canonical motif at position -13 appearing along Diffusion axes 3 in a 3D representation (Fig. S5). This shows that not only can the aSD sequence bind at different distances from the start codon to induce efficient translation initiation, consistent with the interaction neighborhoods shown in Fig. 3d but also that it is hard to evolve a sequence with a shifted SD motif by one or two positions through single point mutations without losing translational efficiency. In contrast, sequences with an SD motif shifted by three positions remain largely connected by extended ridges of functional sequences. In these extended ridges, a second binding site can evolve through a sequence of point mutations without destroying the first. Specifically, within each trinucleotide sequence around the central AGG common to two binding registers, mutations can accumulate in diverse orders, opening up many different evolutionary paths only subject to the constraint of evolving a second SD motif before destroying the first one. Figure 4a shows two examples of such paths highlighted with solid black lines connecting motifs at -14 and -13, with those at -11 and -10, respectively.

Comparing sequence probability across different species

To investigate whether the structure of the genotype–phenotype map is the same across distant species, we performed the same analysis using 5'UTR sequences from 4,328 annotated genes in the genome of *B. subtilis*, whose most recent common ancestor with *E. coli* dates back to ∼2 billion years ago (Feng et al. 1997). We first found that the AG bias marking the location of the SD sequence in the 5'UTR is located approximately 2 bp further upstream from the start codon compared to its location in *E. coli* (Fig. S6A), as previously reported (Hockenberry et al. 2018). We then extracted the nine nucleotides sequences 6 bp upstream of the start codon and inferred the sequence probability distribution using SeqDEFT. The estimated log-probabilities were highly correlated with those obtained from the *E. coli* genome (Spearman $\rho = 0.94$, Fig. S6B), but more importantly, the inferred genotype–phenotype map displayed a similar structure, with peaks corresponding to different binding registers of the aSD sequence and extended ridges connecting sets of sequences with overlapping binding sequences separated by 3 positions (Fig. S6B). Overall, the probability distributions of the SD sequences are quantitatively very similar across distant species and show the same main qualitative features.

Comparing sequence probability and functional measurements

We next compared the genotype–phenotype maps based on genomic sequences with the genotype–phenotype map obtained with MAVE data. First, we directly compared the estimated sequence probability across the *E. coli* genome with the inferred translational efficiency from MAVE data (Fig. 4b) for every possible sequence. We found a moderate nonlinear relationship between these two independently inferred quantities (Spearman $\rho = 0.55$). Sequences with very low estimated probability ($P < 10^{-8}$) consistently showed low translational efficiency ($\log(\text{GFP}) < 1.0$), whereas sequences with high sequence probability ($P > 10^{-4}$) had consistently higher but variable translational efficiencies (mean = 1.84, standard deviation = 0.63).

To investigate whether this modest degree of agreement is due to noise in the estimates for individual sequences or to having inferred qualitatively different genotype–phenotype maps, we applied the visualization technique to the empirical genotype–phenotype map inferred from MAVE data using VC regression (Figs. 4c and S7). Despite the much more skewed phenotypic distribution of estimated translational efficiencies, this low-dimensional representation has essentially the same structure, with isolated peaks corresponding to different distances of the SD motif to the start codon and extended ridges connecting sequences with SD motifs shifted by three positions separated along several Diffusion axes (Figs. 4c and S8). In addition to the previous structure, we identify an additional extended ridge of functional sequences with sequences starting by GAG. This subsequence, together with the upstream G from the fixed genetic context in which the experiment was performed, forms a binding site for the aSD sequence at position -15. In contrast, the probability distribution of SD sequences was inferred from genomic sequences with different flanking nucleotides, such that genotypes starting with GAG, on average, are not as functional. Thus, we can conclude that, despite showing only a moderate quantitative agreement, the two inference procedures using different types of data are able to recover genotype–phenotype maps with the same qualitative features and expected long-term evolutionary dynamics.

Uncertainty quantification for genetic interactions and phenotypic predictions

Visualization of genotype–phenotype maps based on our point estimates enabled the identification of key shared qualitative features and the genetic interactions underlying them. However, we can also evaluate the strength of evidence supporting these interactions by leveraging the uncertainty quantification capabilities of our Gaussian process models as implemented in *gpmap-tools* e.g. we can compute the posterior distribution of the effects of specific mutations in different backgrounds. As an illustration of this strategy, we first validated the incompatibilities separating peaks in the SD genotype–phenotype map by computing the posterior distribution for mutational effects in the two backgrounds UUA**AGGAG**C (gray) and UA**AGGAG**CA (black), which contain the same AGGAG motif shifted by one position (at positions -11 and -10, Fig. 4d). In the absence of epistasis, mutational effects are expected to be exactly the same in the two genetic backgrounds. While this is true for some mutations e.g. C-5A (Fig. 4d), the three mutations that allow shifting the SD motif one position upstream (from -10 to -11) in the UUA**AGGAG**C context (gray) i.e. A-10G, G-8A, and A-7G, are strongly deleterious in that context, but beneficial when introduced in a UA**AGGAG**CA background (black, Fig. 4d). Importantly, the posterior distributions are concentrated around

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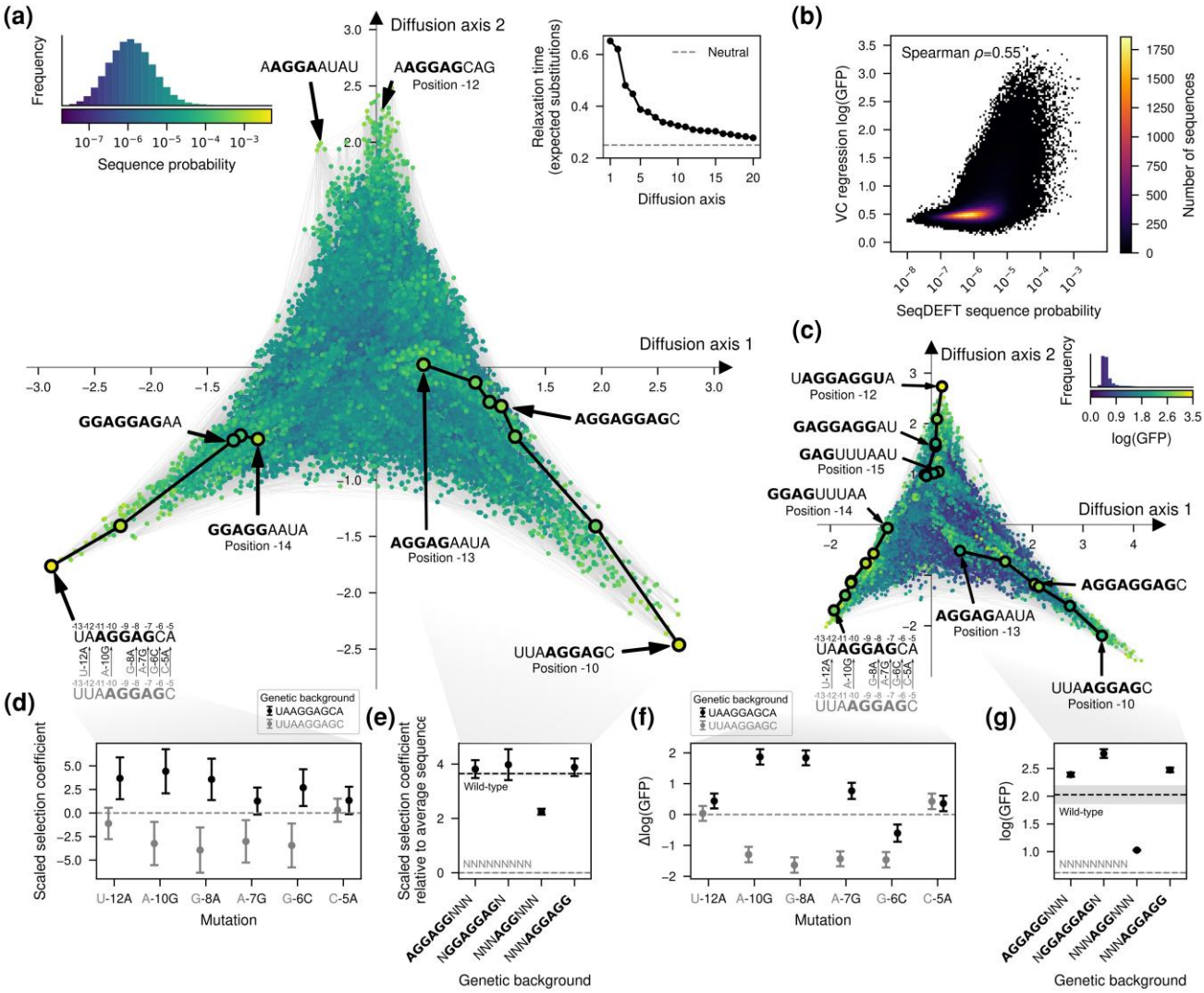

Fig. 4. Visualization of the genotype–phenotype map of the Shine-Dalgarno sequence. a, c) Low-dimensional representation of the *E. coli* Shine-Dalgarno sequence probability distribution inferred with SeqDEFT a) and the translational efficiencies inferred with VC regression (c). Every dot represents one of the 4^9 possible sequences and is colored according to their inferred probability (a) or log(GFP) values (c). The inset represents the distribution of inferred sequence probabilities or log(GFP) values along with their corresponding color in the map. Inset in the upper right corner of a) shows the relaxation times associated to the 20 most relevant Diffusion axes, showing that the first two Diffusion axes have much longer relaxation times than the rest. Sequences are laid out with coordinates given by these first two Diffusion axes and dots are plotted in the order of the third Diffusion axis. b) 2D histogram representing the relationship between the inferred sequence probabilities from their frequency in the *E. coli* genome and the estimated translational efficiencies inferred with VC regression from MAVE data. d) Posterior distribution inferred by SeqDEFT for the scaled selection coefficient of specific mutations when introduced in two genetic contexts, UUAAGGAGC (gray) and UAAGGAGCA (black), representing a shift of the AGGAG motif by one nucleotide. Mutational effects are reported in units of scaled selection coefficients. e) Posterior distribution inferred by SeqDEFT for the average scaled selection coefficient, relative to the average across all possible sequences, for genotypes containing the AGGAGG motif at positions separated by three nucleotides, along with their potential mutational intermediates. f) Posterior distribution inferred with VC regression for the effects of specific mutations on log(GFP) when introduced in two genetic contexts, UUAAGGAGC (gray) and UAAGGAGCA (black), representing a shift of the AGGAG motif by one nucleotide. g) Posterior distribution inferred with VC regression for the average log(GFP) of genotypes containing the AGGAGG motif at positions separated by three nucleotides, along with their potential intermediates. e, g) Horizontal dashed lines represent posterior mean of the average phenotype across all possible sequences (gray) or wild-type sequences, given by the genomic sequences in (e) and AAGGAGGUG in (g) (black). Shaded areas represent the 95% credible intervals. d–g) Points represent the maximum a posteriori (MAP) estimates and error bars represent the 95% credible intervals.

the means, showing that the data strongly supports that mutations needed to shift the SD motif by one position are substantially deleterious in that specific context, creating the valleys that separate the main peaks of this genotype–phenotype map.

We next evaluated the evidence supporting the existence of the extended ridges connecting sequences with an SD motif shifted by three positions. To do so, we computed the posterior distribution for the average phenotype (scaled selection coefficient relative to the average fitness across all sequences) of genotypes containing two overlapping binding registers at positions -13 and -10 (NGGAGGAGN), only one (AGGAGGNNN and NNNAGGAGG), or none (NNNAGGNNN). Whereas NGGAGGAGN, AGGAGGNNN, and NNNAGGAGG are highly functional, sequences with a central AGG have on average roughly half as large of a

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scaled selection coefficient than sequences containing full motifs in either or both registers (Fig. 4e).

The posterior distributions for these same sets of genotypes and mutations estimated from sequence data from the *B. subtilis* genome (Fig. S6D,E) and from VC regression analysis on MAVE data (Fig. 4f,g) are largely concordant. The agreement between these independent data sources, together with uncertainty quantification in each case, provides strong support for a common landscape structure with distinct peaks at the different binding registers of the aSD sequence, connected by extended ridges of functional sequences linking registers offset by three nucleotides in aSD binding position.

A biophysical model recapitulates the qualitative properties of empirical SD genotype–phenotype maps

Although the inferred genotype–phenotype map exhibited extensive epistasis, the visualization revealed that its complexity could be largely explained by a simple underlying mechanism in which the aSD sequence can bind at varying distances from the start codon. We hypothesize that this mechanism alone explains both the existence of isolated peaks and, together with the quasi-repetitive nature of the aSD sequence, the extended ridges. Moreover, despite our ability to estimate mutational effects in different contexts, inference of the actual binding preferences of the aSD from the data is hindered by the convolution of the effects of mutations on the binding affinities at different registers. To tackle these issues, we fit a simple mechanistic model, in which GFP protein abundance is linearly dependent on the fraction of mRNA bound by the aSD at thermodynamic equilibrium at different positions p relative to the start codon, where the binding energy ΔG of the aSD is an additive function of the sequence at that position x_p (Fig. 5a, see Methods). We fit this biophysical model (Fig. S9A) by maximum likelihood to the MAVE dataset and achieved good predictive performance in both training ($R^2 = 0.59$, Fig. S9B) and held-out sequences ($R^2 = 0.64$, Fig. S9C). Importantly, predictions of this simple model for all 4^9 SD sequences recapitulate the main structure of the genotype–phenotype map (Fig. 5c) with isolated peaks, strongly context-dependent mutational effects (Fig. 5e), and extended ridges corresponding to different registers of binding (Fig. 5c,f).

Notably, this model achieves this with only 27 free parameters. But more importantly, these parameters have clear biophysical interpretations e.g. in terms of mutational effects on binding energies. This allowed us to deconvolve the effects of mutations on binding at different registers and to infer the allele and position specific energetic contributions to binding (Fig. 5b). As expected, the reverse complement of the aSD is the most stable binder, but different mutations have substantially variable effects in the binding energy. Not only do some positions have stronger energetic contributions in general (positions 2–5 within the SD sequence), but different mis-smatches with the aSD in the same position have different energetic effects e.g. A4G is only slightly destabilizing ($\Delta\Delta G = 0.84$ kcal/mol), whereas A4C is highly destabilizing ($\Delta\Delta G = 1.98$ kcal/mol). Moreover, we can verify that the peaks correspond to different binding registers by computing the binding energy of every sequence at specific positions relative to the start codon and color the visualization by those energies (Fig. 5d). Similarly, this representation also shows that the context-dependent mutational effects across peaks can be explained by the differences in binding energies across different positions (Fig. 5g) and that the extended ridges of functional sequences correspond to sequences that can be strongly bound at positions separated by three nucleotides (Fig. 5d,h).

We next compared our thermodynamic model to an alternative thermodynamic model based on base-pair stacking interaction energies from classical RNA folding algorithms (Salis et al. 2009; Lorenz et al. 2011). Interestingly, our model achieved higher predictive accuracy (Fig. S10A,B, $R^2 = 0.44$) and more faithfully captured the overall structure of the empirical genotype–phenotype map (Fig. S10C). These findings suggest that the SD:aSD interaction *in vivo* cannot be fully explained by RNA thermodynamics alone, highlighting the importance of molecular context in shaping RNA–RNA interactions. More broadly, this analysis shows how visualization can guide the construction of simplified, biophysically interpretable models that reproduce the key qualitative features of genotype–phenotype maps.

Discussion

In this paper, we present *gpmap-tools*, an extensively documented software library with tools for the inference, visualization and interpretation of empirical genotype–phenotype maps containing arbitrarily complex higher-order genetic interactions. By providing a framework for the analysis of complex genetic interactions, *gpmap-tools* has the potential to reveal the simple qualitative properties of these complex mappings and to aid in development of biophysical and mechanistic hypotheses for these observed features.

The first step in this framework is the inference of the complete genotype–phenotype map comprising all possible sequences from either experimental MAVE data or sequence counts. Taking into consideration the noise in the data (due either to sampling noise or experimental error), *gpmap-tools* is capable of computing the high-dimensional posterior distribution over all possible genotype–phenotype maps under a variety of priors. This allows us to obtain the MAP estimate, that is, the most probable genotype–phenotype map given the observed data and prior distribution. However, in contrast to other expressive models able to capture complex genetic interactions, such as neural networks (Bryant et al. 2021; Gelman et al. 2021; Sethi and Zhou 2025), our inference methods provide rigorous uncertainty quantification of phenotypes, mutational effects or any linear combination of phenotypic values. This is important, as it tells the user which phenotypic predictions, mutational effects or genetic interactions can be trusted and to what extent, given the data.

The second step in this framework is the interpretation of the inferred genotype–phenotype maps. *gpmap-tools* provides a powerful method for visualizing fitness landscapes (McCandlish 2011) that allows exploratory data analysis, interpretation and comparison of complex datasets and models. Thus, rather than interpreting the results through an explicit parametric model allowing higher-order genetic interactions (Poelwijk et al. 2016; ; Faure and Lehner 2024; Faure et al. 2024; Park et al. 2024) or descriptive statistics like the number of peaks or adaptive walks (Szendro et al. 2013; Ferretti et al. 2018; Papkou et al. 2023; Westmann et al. 2024b; Chattopadhyay et al. 2025; Li and

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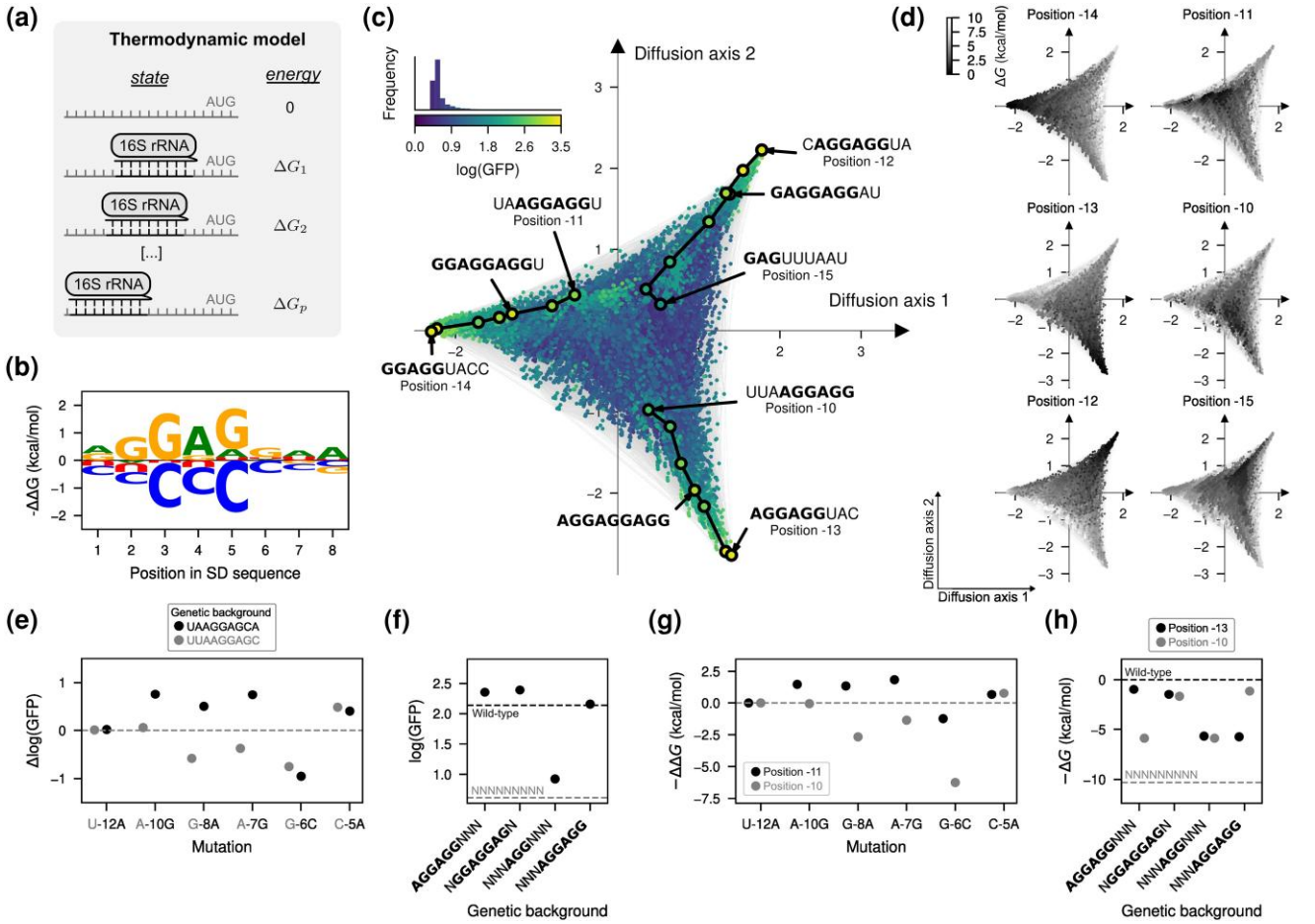

Fig. 5. Biophysical model of sequence-dependent translational efficiency. a) Thermodynamic model of binding of the 16S rRNA 3′ tail to the 5′ UTR of mRNAs at different positions *p* relative to the start codonAUG. b) Sequence logo representing the site-specific but register-independent allelic contributions to the binding energy, where the size of the letter represents the difference in binding energy to the average across nucleotides. c) Visualization of the genotype–phenotype map that results from predicting the phenotype of every possible sequence under the inferred thermodynamic model. Every dot represents one of the 4^9 possible sequences and is colored according to the predicted log(GFP). The inset represents the phenotypic distribution along with their corresponding color in the map. Sequences are laid out according to the first two Diffusion axes and dots are plotted in order according to Diffusion axis 3. d) Visualization of the genotype–phenotype map under the inferred thermodynamic model representing the binding energies at positions -15 to -10 relative to the start codon showing that the peaks in the visualization correspond to the strongest binding at different positions and extended ridges correspond to sequences that are bound in two registers separated by three nucleotide positions. Dots are plotted in reverse order of binding energy in the corresponding register. e,g) Effects of specific mutations on log(GFP) e) and binding energies at positions -11 and -10 g) when introduced in two genetic contexts, UUAAGGAGC (gray) and UAAGGAGCA (black), representing a shift of the AGGAG motif by one nucleotide. f,h) Thermodynamic model's predictions for the average log(GFP) (f) and binding energies at positions -13 and -10 h) of genotypes containing the AGGAGG motif at positions separated by three nucleotides, along with their potential intermediates. Horizontal dashed lines represent posterior mean of the average phenotype across all possible sequences (gray) or wild-type sequences, given by AAGGAGGUG in f) and AGGAGGAA in h), (black). (b,d,g,h) Binding energies are reported in units of kcal/mol assuming a temperature of 37°C and are relative to the strongest binding sequence AGGAGGAA (as inferred from (b)) at any register under the inferred model.

Zhang 2025), this method leverages the evolutionary dynamics on the genotype–phenotype map to highlight its main, potentially unexpected, qualitative features. Thus, we can use the visualization to generate hypotheses for how mutational effects change across genetic backgrounds and test these predictions by computing the corresponding background-dependent posterior distributions (Figs. 4 and S6). Identifying the main features of the genotype–phenotype map can be crucial for defining an appropriate mechanistic or biophysical model. Here, visualization of the Shine-Dalgarno landscapes allowed us to define a thermodynamic model in which the binding energy depends only additively on the sequence at each register, and to verify that this simple model recapitulated the main qualitative features of the landscape (Fig. 5). Additionally, this technique enabled a detailed comparison of genotype–phenotype maps inferred with different methods and data sources. In contrast to broadly used metrics, like Pearson or Spearman coefficients, this method shows the extent to which different landscapes have the same structure and qualitative features. In this study, it showed that genotype–phenotype maps inferred from two distantly related species *E. coli* and *B. subtilis* (Figs. 4 and S6), as well as from entirely independent data sources (MAVE experiments versus

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natural sequence data), exhibited strikingly similar structure despite only moderate quantitative agreement (Fig. 4). Identifying consistent structures across different data types and sources is essential for linking experimentally measured landscapes to the evolutionary forces shaping regulatory sequences, given that true fitness values in natural populations are typically unknown. More broadly, our visualization technique enables comparison of genotype–phenotype maps across different classes of genetic elements, such as regulatory sequences, protein–protein interactions and enzymes, by revealing shared landscape features that may reflect similar evolutionary dynamics, despite differences in biological context.

The methods implemented in *gpmap-tools* scale to genotype–phenotype maps with millions of sequences by making several modeling assumptions, which also entail certain limitations. First, MEI and VC regression are phenomenological models. As such, they do not explicitly model global or nonspecific epistasis that often arises from nonlinear dependencies between the underlying quantities affected by mutations and our measurements (Bloom 2015; Otwinowski et al. 2018; Tonner et al. 2022; Faure and Lehner 2024). Instead, these models rely on learning the pervasive genetic interactions induced by these global nonlinearities to nonetheless make accurate phenotypic predictions. Second, SeqDEFT assumes that observed sequences are drawn independently from the underlying probability distribution. While this assumption may hold for a few specific regulatory sequences that are repeated many times along the genome of a single species e.g. the Shine-Dalgarno sequence or the 5' splice site (Chen et al. 2021), it remains unclear how robust it is to the known challenge of using phylogenetically related sequences from widespread multiple sequence alignments of protein families (Hockenberry and Wilke 2019; Rodriguez Horta and Weigt 2021; Dietler et al. 2023). Third, both inference and visualization methods still require storing all possible sequences and their phenotypes in memory. The number of such sequences grows exponentially with sequence length, limiting the applicability of *gpmap-tools* to spaces of sequences of a constant and relatively short length (5 amino acids, 12 nucleotides, 24 biallelic sites). Despite these limitations, *gpmap-tools* provides a unique set of tools for studying the genotype–phenotype maps of short genetic elements. By combining nuanced analysis of epistasis, rigorous uncertainty quantification, and the capacity to infer landscapes containing millions of genotypes, it serves as a necessary stepping stone towards understanding the vastly larger genotype–phenotype maps arising at the gene, protein, and genome-wide scale.

Methods

Sequence diversity of the Shine-Dalgarno sequence

We downloaded the *E. coli* genome and annotation from Ensembl bacteria release 51, built on assembly version ASM160652v1, and *B. subtilis* assembly ASM904v1 from GeneBank. We extracted the 5'UTR sequence for every annotated gene using *pysam* (Li et al. 2009; Bonfield et al. 2021) and kept the 5,311 and 4,328 sequences, respectively, for which we could extract 20 bp upstream of the start codon without any ambiguous character "N". These sequences were aligned with respect to the start codon and used for computing site-frequency logos using *logomaker* (Tareen and Kinney 2020) and estimating the complex probability distribution using the *gpmap-tools* implementation of SeqDEFT (Chen et al. 2021). The MAP estimate was used to compute the coordinates of a low-dimensional representation assuming that the stationary distribution of the evolutionary random walk matches the estimated sequence probabilities by selecting a proportionality constant of $c = 1$ and uniform mutation rates.

Analysis of the experimental fitness landscape of the Shine-Dalgarno sequence

Phenotype data were computed from the processed data for independent replicates conducted in the dmsC genetic background as reported in the original manuscript (Kuo et al. 2020). The mean and standard error was computed for all the 257,565 measured sequences. We estimated a common measurement variance of $\hat{\sigma}^2 = 0.058$ using genotypes measured across all three experimental replicates. The squared standard error for each genotype i was computed by dividing the overall experimental variance $\hat{\sigma}^2$ by the number of replicates n_i in which each sequence was measured ($\hat{\sigma}_i^2 = \hat{\sigma}^2/n_i$). We kept 0.1% of the sequences as test set, and use the remaining sequences for fitting different models to infer the complete genotype–phenotype map while evaluating their performance on the held-out test data. We estimated the variance components from the empirical distance-correlation function and used them to define a Gaussian process prior for inference of the complete combinatorial landscape containing all 4^9 genotypes, taking into account the known experimental variance $\hat{\sigma}_x^2$ for every sequence. We also computed the posterior mean and variances across all test sequences to assess the accuracy of the predictions and the calibration of the posterior probabilities in held-out data. We used the MAP estimate to compute the coordinates of the visualization assuming several different average values of log(GFP) under the stationary distribution that ranged from 1 to 2.5 (Fig. S7). An average log(GFP) of 2 at stationarity was selected and used for all subsequent visualizations, similar to our MAP estimate of a log(GFP) of 2.03 for the wild-type reference.

Thermodynamic model of the Shine-Dalgarno genotype–phenotype map

We assume that translation is limited by the initiation step, which is itself modulated by the binding of the 16S rRNA to the 5'UTR of the mRNA, where we assume that the mRNA concentration is independent of the identity of the Shine-Dalgarno sequence. Binding and dissociation are assumed to be much faster than the rate at which translation is effectively initiated, so that the protein abundance is proportional to the fraction of mRNA bound by the 16S rRNA across all registers p at thermodynamic equilibrium, where we assume that binding occurs in at most one register at a time. The fraction of mRNA bound at thermodynamic equilibrium depends on the binding energy ΔG of the 16S rRNA to the mRNA to the sequence x_p starting at each position p, the temperature, which is assumed to the 37°C (310K), and the universal gas constant $R = 1.9872 \times 10^{-3}$ kcal/mol K^{-1}.

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The overall GFP concentration for a sequence x depends on the fraction of bound mRNA and the translation rate when bound β:

$$[\mathrm{GFP}](x) = \beta\left(\frac{\sum_p e^{-\frac{\Delta G(x_p)}{RT}}}{1+\sum_p e^{-\frac{\Delta G(x_p)}{RT}}}\right), \quad (1)$$

where $\Delta G(x_p)$ is the energy of binding of the 16S rRNA to the 8-nucleotide subsequence x_p at position p. The binding energy ΔG is independent of the position p at which binding occurs relative to the start codon and depends additively on the sequence x_p alone given by $\Delta G(x_p) = \Delta G_0 + \sum_i \sum_c x_p(i, c)\Delta\Delta G_{i,c}$, where $x_p(i, c)$ takes value 1 if sequence x_p has allele c at position i and 0 otherwise, ΔG_0 represents the average binding energy across every possible sequence and $\Delta\Delta G_{i,c}$ is the energetic contribution of allele c at position i, subject to the constraint $\sum_c \Delta\Delta G_{i,c} = 0$ across all positions i. In order to incorporate the effect of mutations in binding registers spanning both fixed and variable regions of the sequence, we extended the variable 9 nucleotide sequences with the fixed upstream and downstream sequences CCG and UGAG from the dmsC genetic context.

Following previous work (Kuo et al. 2020), we assume that occupancy at thermodynamic equilibrium is low so that $\sum_p e^{-\frac{\Delta G(xp)}{RT}} \ll 1$, and thus $[\mathrm{GFP}](x) \approx \beta \sum_p e^{-\frac{\Delta G(xp)}{RT}}$. We also model a background fluorescence signal β_0 due to cells autofluorescence in the GFP channel even in the absence of GFP, which is independent of the variable 5'UTR sequence in the experiment. Finally, we consider that experimental errors lie on the log-scale, such that the measured log(GFP) y for sequence x is observed with known noise variance $\hat{\sigma}_x^2$ and an extra or uncharacterized variance σ^2 under a Gaussian likelihood function given by

$$\mathrm{p}(y \mid x) = N\left(\mu_x, \hat{\sigma}_x^2 + \sigma^2\right), \quad (2)$$

where μ_x is the expected log(GFP) under the model given by

$$\mu_x = \log\left(\beta_0 + \sum_p e^{-\frac{\theta_0 + \sum_i \sum_c x_p(i,c)\Delta\Delta G_{i,c}}{RT}}\right) \quad (3)$$

and $\theta_0 = RT\beta + \Delta G_0$. We used PyTorch to encode the model and used the Adam optimizer with a learning rate of 0.02 for 1500 iterations, while monitoring for convergence (Fig. S9A), to find the maximum likelihood estimates of the model parameters. Here, we estimated a $\hat{\beta}_0 = 0.47$ representing the background fluorescence in the absence of aSD binding to the 5'UTR. For estimates of the other parameters see Results and Fig. 5.

Additionally, we fit a 4-parameter calibration model using ensemble binding energies ΔG_x computed with a thermodynamic model of RNA folding and interaction (Lorenz et al. 2011), where $\mu_x = \log\left(\beta_0 + e^{\theta_0 + \theta\Delta G_x}\right)$. Specifically, we used RNAcofold v2.4.9 with -p0 option for computing the binding energy of each SD variant, embedded between the CCG and UGAG flanking sequences, with the anti-SD sequence ACCUCCU across all possible binding configurations. Several candidate anti-SD sequences, ranging from 5 to 9 nucleotides, were tested; ACCUCCU was selected due to the higher predictive power of GFP abundance of the resulting model.

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Supplementary material

Supplementary material is available at *Molecular Biology and Evolution* online.

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Conflicts of interest

None declared.

Data availability

gpmap-tools is an open-source library with source code available at https://github.com/cmarti/gpmap-tools. It is thoroughly documented with several tutorials and explanations of the provided functionalities at https://gpmap-tools.readthedocs.io. Code and data required to reproduce the analyses of the Shine-Dalgarno landscapes is available at https://doi.org/10.5281/zenodo.18500314 and https://github.com/cmarti/shine_dalgarno.

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